

MAJOR PROJECT 2025-2026



B Tech CSE-AIML IV Year II Semester

Department of CSE- Artificial Intelligence and Machine Learning

Major Project Title : KisanSathi: AI-Assisted Farmer Support System

Name of the Internal Guide : Ms. S. Pravallika

Batch Number: 19	
Student Name	Roll Number
Manvitha Pola	22311A6671
Koyyalamudi Rohit Kumar	22311A6670
S. Aashrith Reddy	22311A66C0

ABSTRACT

KisanSathi is an AI-assisted platform that connects farmers directly with government authorities and agricultural experts, providing verified crop prices, scheme updates, expert guidance, and multilingual resources. It also offers AI-based disease detection, yield prediction, fertilizer recommendations, and farmer collaboration, reducing reliance on middlemen and supporting informed, sustainable farming. By centralizing information and expert support in a single platform, the system enhances transparency and improves agricultural decision-making. Ultimately, *KisanSathi* aims to empower farmers with reliable data and digital tools that can improve productivity and income.

EXISTING SYSTEM

- Farmers depend on middlemen for crop prices, leading to exploitation and misinformation.
- Government portals and apps exist, but information is scattered and not farmer-friendly.
- Limited access to expert guidance; responses are slow and not personalized.
- Existing platforms lack unified AI-driven insights (disease detection, yield prediction).
- Poor farmer-to-farmer collaboration and knowledge sharing.
- Language barriers limit access to online agricultural resources.

PROPOSED SYSTEM

- **Role-Based Access:** Farmers access resources and AI tools; Experts provide guidance; Government admins update prices and schemes.
- **Verified Crop Prices:** Official rates updated directly by government authorities, reducing middlemen influence.
- **AI-Driven Support:** Crop disease detection, yield prediction, fertilizer recommendations, and price forecasting.
- **Farmer Collaboration:** Connects farmers growing similar crops to share experiences and discuss challenges.
- **Expert Insights:** Researchers upload tutorials, best practices, and respond to farmer queries for scientific guidance.
- **Multilingual Knowledge Hub:** Blogs, guides, and tutorials available in local languages for better accessibility.

ADVANTAGES

- **All-In-One Platform:** Combines pricing, AI insights, expert support, schemes (*vs. AgriApp, mKisan, Kisan Hub — separate features*)
- **Verified Government Data:** Prices updated by officials (*vs. AgriMarket, Farmers Friend — user-reported/estimated*)
- **AI-Based Decisions, Not Just Information:** Disease detection & yield prediction (*vs. KVK, AgriApp — advisory only*)
- **Role-Based Access:** Farmers, Experts, Government roles (*most existing apps focus only on farmers*)
- **Farmer Collaboration Network:** Connect farmers with same crop (*rare or missing in current platforms*)
- **Multilingual Accessibility:** Local language support (*many platforms only regional or English-centric*)

TECHNOLOGY STACK

- **Frontend:** React.js, HTML, TailwindCSS
- **Backend:** Flask (Python)
- **Database:** MySQL
- **AI Frameworks:** TensorFlow, PyTorch, scikit-learn, OpenCV
- **Chatbot Integration:** OpenAI API
- **Hosting & Deployment:** Render
- **Version Control:** GitHub

CONCLUSION

- KisanSathi connects farmers with government authorities and experts through a unified, reliable platform.
- It reduces middlemen dependency by providing verified crop prices and official agricultural updates.
- AI features such as disease detection, yield prediction, and fertilizer recommendations support informed farming decisions.
- The platform enables collaboration among farmers, encouraging shared learning and collective problem-solving.
- Multilingual access and simple design make it suitable for diverse farming communities.
- Overall, KisanSathi promotes transparency, boosts productivity, and advances digital agriculture.

REFERENCES

1. L. Madaan, A. Sharma, P. Khandelwal, S. Goel, P. Singla and A. Seth, “Price forecasting & anomaly detection for agricultural commodities in India,” in Proc. 2019 ACM Int. Conf. on Computing and Communication Systems (COMPASS ’19), Accra, Ghana, Jul. 3-5, 2019. doi:10.1145/3314344.3332488.
2. A. S. Doutoum and B. Tugrul, “A Review of Leaf Diseases Detection and Classification by Deep Learning,” IEEE Access, vol. XX, 2023. doi:10.1109/ACCESS.2023.3326721
3. Idrees, M., “Assessing the Role of the Middle Man and Information and Communication Technologies (ICTs) in Agriculture Marketing in District Swat.” BS Economics Thesis, Department of Economics & Development Studies, University of Swat, Pakistan, Fall 2013–2017. Jan. 03, 2023.
4. Y. Wang, “Agricultural Products Price Prediction Based on Improved Hybrid Forecasting Framework,” *Crop & Pasture Science*, 2023. DOI:10.1080/08839514.2023.2204600.

THANK YOU